AgentBoard: An Analytical Evaluation Board of Multi-turn LLM Agents

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Abstract

Evaluating Large Language Models (LLMs) as general-purpose agents is essential for understanding their capabilities and facilitating their integration into practical applications. However, the evaluation process presents substantial challenges. A primary obstacle is the benchmarking of agent performance across diverse scenarios within a unified framework, especially in maintaining partially-observable environments and ensuring multi-round interactions. Moreover, current evaluation frameworks mostly focus on the final success rate, revealing few insights during the process and failing to provide a deep understanding of the model abilities. To address these challenges, we introduce AGENTBOARD, a pioneering comprehensive benchmark and accompanied open-source evaluation framework tailored to analytical evaluation of LLM agents. AGENTBOARD offers a fine-grained progress rate metric that captures incremental advancements as well as a comprehensive evaluation toolkit that features easy assessment of agents for multi-faceted analysis. This not only sheds light on the capabilities and limitations of LLM agents but also propels the interpretability of their performance to the forefront. Ultimately, AGENTBOARD serves as a step towards demystifying agent behaviors and accelerating the development of stronger LLM agents.[1]

1 Introduction

General-purpose agents that can autonomously perceive and act in various environments are considered significant milestones in Artificial Intelligence (Russell and Norvig, 2005). Recent advancements in large language models (OpenAI, 2023; Touvron et al., 2023) have demonstrated emergent agent abilities that enable them to understand diverse environments and perform step-by-step planning through multi-round interactions (Yao et al., 2023; Song et al., 2023). These advanced abilities contribute to the potential of LLMs to act as generalist agents for real-world problem-solving.

A comprehensive evaluation of LLM agents is crucial for the progression of this emerging field. To start, *task diversity* is necessary to cover various agent tasks such as embodied, web, and tool agents. Additionally, *multi-round interaction* is critical to mimic realistic scenarios, in contrast to the single-round tasks commonly adopted in existing benchmarks (Xu et al., 2023b; Lin and Chen, 2023; Qin et al., 2023a). Furthermore, evaluating agents in *partially-observable* environments, where they must actively explore to understand their surroundings, is essential for practical assessments. This differs from the "synthetic" agent tasks (Wang et al., 2023b) derived from conventional benchmarks in fully-observable environments, such as MMLU (Lanham et al., 2023) and GSM8K (Cobbe et al., 2021). However, existing agent benchmarks fail to satisfy all of these criteria.

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[1]Code and data are available at https://github.com/hkust-nlp/AgentBoard

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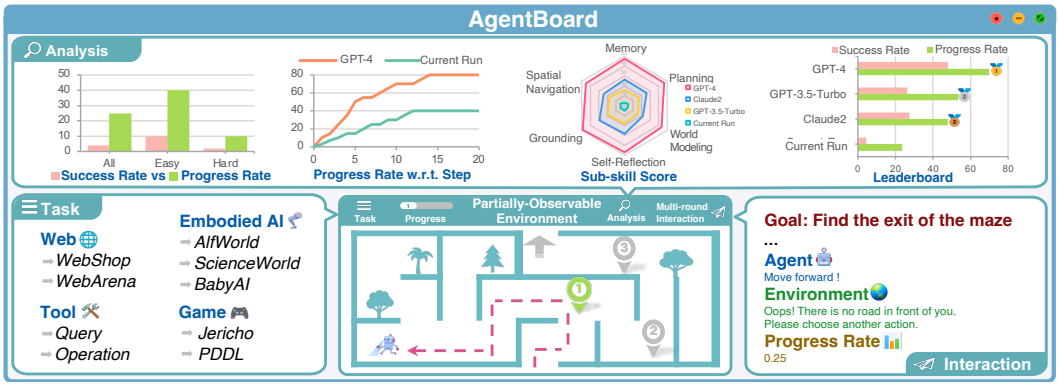

Figure 1: The illustrative overview of AGENTBOARD. AGENTBOARD consists of a 9 diverse tasks. Agents interact in multi-rounds with partially-observable environments to achieve each subgoal. Furthermore, AGENTBOARD provides an open-source analytical evaluation framework, as shown in the figure.

Moreover, the inherent complexity in agent tasks characterized by multi-round interactions distinguishes them significantly from other language tasks. Due to this complexity, there is a pressing need to delve into the details and gain a deeper understanding of how models function during the process. Nonetheless, most current evaluations predominantly rely on the final success rate as their metric, which provides limited insights into these intricate processes (Liu et al., 2023a; Wang et al., 2023b; Yao et al., 2023; Liu et al., 2023b; Mialon et al., 2023). This simplified evaluation is particularly inadequate in challenging environments where most models demonstrate nearly zero success rates, consequently blurring finer distinctions and obscuring underlying mechanisms (Liu et al., 2023a).

To address these issues, we introduce AGENTBOARD, a benchmark designed for multi-turn LLM agents, complemented by an analytical evaluation board for detailed model assessment beyond final success rates. AGENTBOARD encompasses a diverse set of 9 unique tasks and 1013 exemplary environments, covering a range from embodied AI and game agents to web and tool agents. Each environment, whether newly created or adapted from pre-existing ones, is carefully crafted and authenticated by humans to ensure multi-round and partially observable characteristics in a unified manner. Notably, we have defined or manually annotated subgoals for each data sample, introducing a unified *progress rate* metric to track the agents' detailed advancements. As we will demonstrate in §4.2, this metric uncovers significant progress made by models that would otherwise appear trivial due to negligible differences in success rates.

Along with the benchmark, we develop the AGENTBOARD evaluation framework as an open-source toolkit that features an analytical web panel to examine various dimensions of agent abilities through interactive visualization. The toolkit offers a unified interface, providing users with easy access and effortless customization options. As shown in Figure 1, the AGENTBOARD toolkit currently supports analysis and visualization on fine-grained progress rates tracking, performance breakdown for hard and easy examples, detailed performance across various sub-skills, long-range interaction assessment, grounding accuracy, and trajectory. This detailed evaluation is crucial for acknowledging the progress of LLM agents and for guiding the development of more robust LLM agent models. The comparison between AGENTBOARD and previous works is shown in Table 1.

We evaluated a range of proprietary and open-weight LLM agents using AGENTBOARD, obtaining insights into the current landscape of LLMs as agents. Key findings include: (1) GPT-4, unsurprisingly, outperforms all other models by exhibiting extensive proficiency across distinct agentic abilities with Llama3 (Touvron et al., 2023) and DeepSeek LLM (DeepSeek-AI et al., 2024) taking the lead; (2) Strong LLM agents are characterized by their capability for *multi-turn* interaction with the environment, an ability that is notably lacking in most open-weight models; (3) Emergent agentic abilities are strongly dependent on basic abilities like grounding, world modeling, and self-reflection. Current proprietary models typically demonstrate comprehensive agentic abilities, while open-weight LLMs show varying deficiencies. Through AGENTBOARD, we highlight the importance of analytic evaluation of LLM agents. The detailed evaluations provided by AGENTBOARD and its open-source toolkit are expected to significantly contribute to the further development of LLM agents.

Table 1: AGENTBOARD differs from other LLM benchmarks by providing a comprehensive framework that integrates all four guiding principles within its evaluation system. [z]Notably, AgentBench entails both single and multi-round tasks, with mainly the former differentiating open-sourced models. [x]The GAIA benchmark focuses solely on question answering tasks. [y]MINT benchmark primarily includes fully-observable environments tasks derived from conventional evaluations such as HumanEval and GSM8K.

Benchmarks	Task Diversity	Multi-round Interaction	Partially-Observable Environments	Fine-grained Progress Metrics	Analytical Evaluation
AgentBench (Liu et al., 2023a)	4	7 [z]	4	7	7
GAIA (Mialon et al., 2023)	7 [x]	4	4	7	7
MINT (Wang et al., 2023b)	4	4	7 [y]	7	7
API-Bank (Li et al., 2023)	7	4	4	7	7
ToolEval (Qin et al., 2023b)	7	4	4	7	7
LLM-Eval (Lin and Chen, 2023)	4	7	7	7	7
AGENTBOARD	4	4	4	4	4

2 AGENTBOARD – Overview

AGENTBOARD is a uni ed, open-source benchmark for evaluating LLM agents that adheres to ve key principles: task diversity, multi-round interaction, partially-observable environments, ne-grained metrics, and analytical evaluation, as shown in Table 1. Our commitment to these principles manifests in three key areas:

- Task Diversity and Uniformity, where we carefully curate nine diverse environments across four scenarios, ensuring they require multi-round interactions, are fully text-based and primarily partially-observable. This contrasts with many existing LLM agent benchmarks, which are often solvable in a single round, derived from fully-observable tasks like MMLU, or focus on a speci c type of task, as demonstrated in Table 1. Compared to AgentBench (Liu et al., 2023a), in particular, our choice of tasks makes AGENTBOARD planning-heavier, where the results on AGENTBOARD well-correlates with the scores on traditional reasoning and coding benchmarks, while the AgentBench scores strongly correlated with scores on the knowledge test MMLU, as illustrated in Ruan et al. (2024). We elaborate on the choice of tasks and their adaptation in §3.
- Fine-grained Progress Rate, where AGENTBOARD is the rst to propose a ne-grained progress rate metric tracking the intermediate progress of different agents. This metric distinguishes our benchmark in tracking minimal improvement in LLM agent performances. Such a capability is crucial in current endeavors to develop stronger open-weight LLMs, providing detailed insights that are essential for incremental advancements in agent capabilities. We provide detailed introduction for this metric and its annotation process in §2.2.
- Comprehensive Analysis, where AGENTBOARD is the rst LLM Agent benchmark to expand metrics beyond mere success rate and scores to include detailed analyses. As illustrated in Figure 1, such a comprehensive evaluation includes (1) ne-grained progress rates tracking different agents, (2) grounding accuracy, (3) performance breakdown for hard and easy examples, (4) long-range interactions, (5) analyses of performance across various sub-skills, and (6) trajectory with friendly visualization. We elaborate these analyses in our experiments at §4. Additionally, AGENTBOARD provides an web interface [2] through Wandb dashboard that offers interactive visualizations of these analyses during evaluation. We perform a case study on the panel in §5.

2.1 A Uni ed Multi-Round Re ex Agent

Preliminaries: An LLM agent receives textual world descriptions, chooses a text action, and gets feedback detailing state changes and any action errors. Interaction with these environments can be modeled as a special case of Partially Observable Markov Decision Processes (POMDPs) de ned by tuple $\langle g, S, A, O, T \rangle$, with goal g, state space S, valid actions space A, observation space (including environment feedback) O, transition function $T : S \times A \to S$. An agent with policy π makes prediction at time step t based on goal g and memory $m_t = f(o_j, a_j, o_{j+1}, a_{j+1}, \ldots, o_t | g), 0 \le j < t$, which is a sequence of actions and observations. This trajectory of the agent $\tau = [s_0, a_0, s_1, a_1, \ldots, s_t]$ is formulated by policy and environmental state transitions, such as

$$p_\pi(\tau) = p(s_0) \prod_{t=0}^{T} \pi(a_t | g, s_t, m_t) T(s_{t+1} | s_t, a_t) \quad (1)$$

[2] An example of the panel is public here.